MATH 321

SOCIAL MEDIA USAGE & MENTAL HEALTH

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INTRODUCTION

- Social media has become a dominant part of modern communication
- It helps people stay connected, but it also may influence our mental health
- Researchers are increasingly studying how things like screen time, content exposure, and online interactions affect stress, sleep, and happiness
    - One literature review found that daily social media use is associated with increased stress, anxiety and poorer sleep quality — and also lower life satisfaction and mood

OUR DATA

CONTINUOUS VARIABLES:

Age: 16–49

16–49

Mean = 33.0)

Daily Screen Time:
1.0–10.8 hrs

1.0–10.8 hrs

Mean = 5.53)

Sleep Quality:

2–10

Mean = 6.30)

Days Without Social Media:

0–9

Mean = 3.13)

Exercise Frequency:
0–7 times/week

0–7

Mean = 2.45)

Happiness Index:

4–10

Mean = 8.38)

CATEGORICAL VARIABLES:

Gender:
500 entries

Primary Social Media Platform:
500 entries

SUMMARY STATISTICS:

Age (years)

- Mean = 32.99
- SD = 9.96
- Range: 16-49

Screen Time (hours)

- Mean = 5.53
- SD = 1.73
- Range: 1.00-10.8

Sleep Quality

- Mean = 6.304
- SD=1.53
- Range: 1-10 scale

Days without Social Media

- Mean = 3.134
- SD = 1.86
- Range: 0-9

Exercise frequency

- Mean = 2.448
- SD = 1.43
- Range: 0-7 scale

Happiness Index

- Mean = 8.38
- SD = 1.52
- Range: 1-10 scale

Number of Responses per Group

Preferred Social Media Platform

- Facebook: 81
- Instagram: 74
- LinkedIn: 87
- TikTok: 95
- X (Twitter): 88
- YouTube: 75

Gender

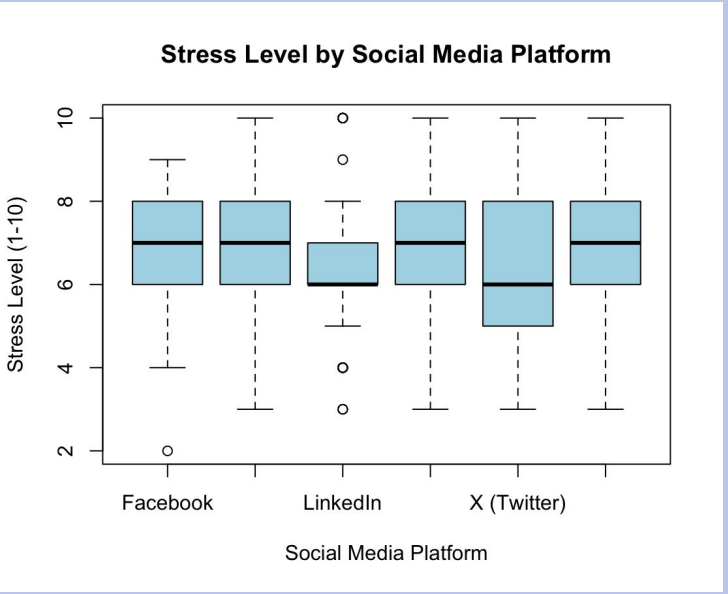
- Female: 229
- Male: 248
- Other: 23

RESEARCH QUESTIONS & HYPOTHESIS

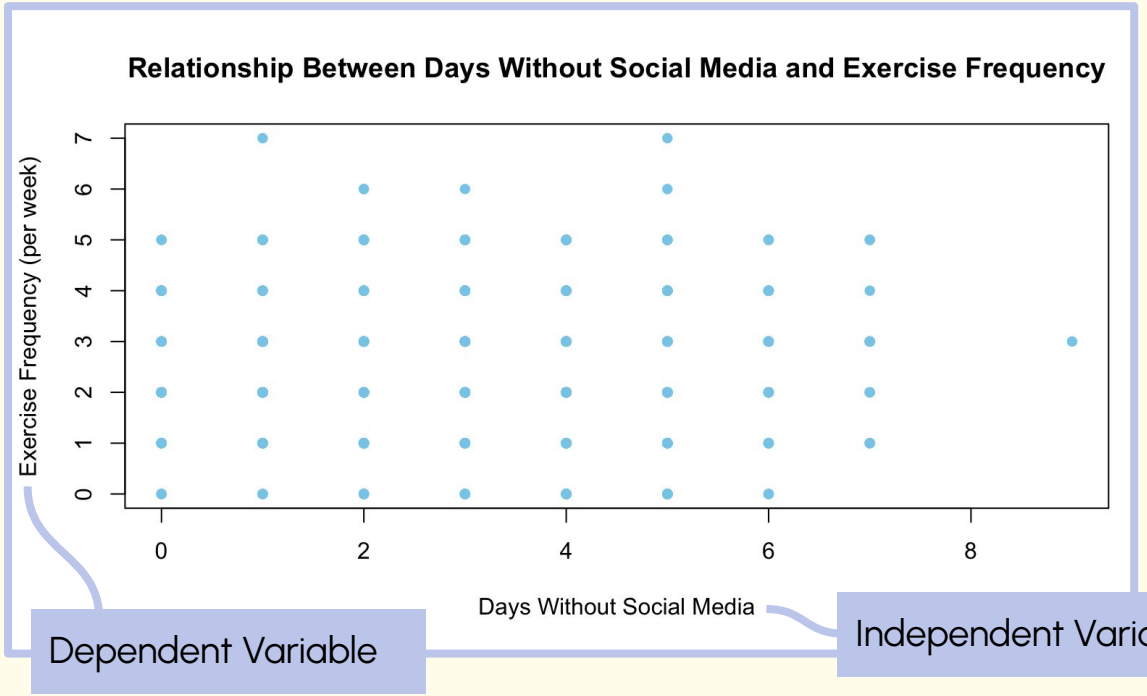
- **BIG PICTURE QUESTION:** How does social media use impact mental health and well-being?
- **OVERALL HYPOTHESIS**: Increased time spent on social media is directly associated with poor mental health outcomes, specifically higher stress levels, lower sleep quality, and a lower happiness index.
- **RQ1**: Does stress level differ across primary social media platforms?
- **RQ2**: Does exercise frequency differ based on the number of days without social media?
- **RQ3**: Does exercise frequency differ based on the number of days without social media?
- **RQ4**: Is there a relationship between daily screen time and happiness index?

RQ1: STRESS LEVELS BY PLATFORM

- **Independent Variable**: Primary social media platform (TikTok, X, LinkedIn, Facebook, YouTube, Other)
- **Dependent Variable**: Stress level (1–10)
- Used a one-way ANOVA to test whether average stress levels differ across primary social media platforms
- $F(5, 494) = 1.13$, $p = 0.342$
- **Not significant** → no differences in mean stress levels across platforms
- Boxplot shows distribution of stress levels for each platform



RQ2: EXERCISE FREQUENCY BY DAYS WITHOUT SOCIAL MEDIA

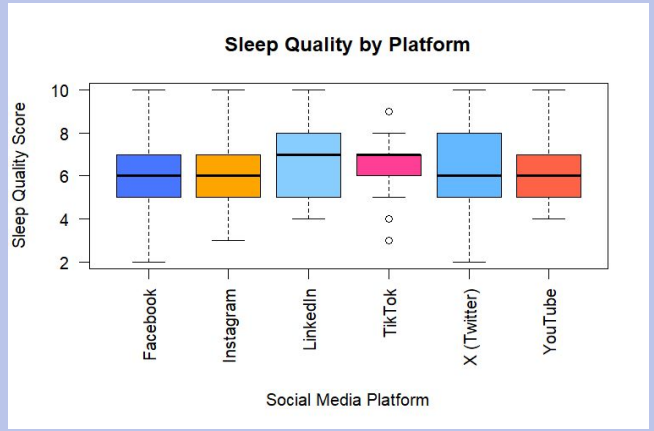


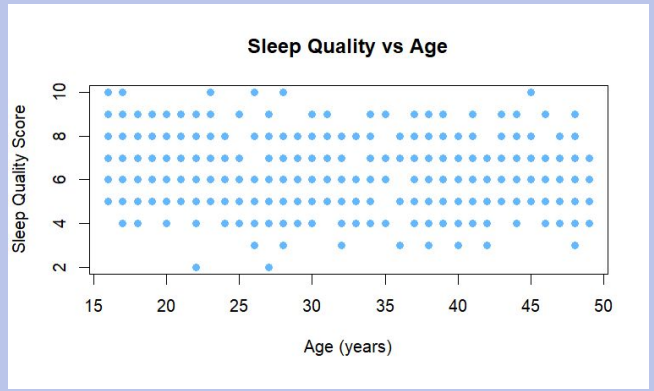
- Used a simple linear regression
- Testing whether days without social media predict exercise frequency
- t = 0.00, f = 0.00000007267, p = 1.00

Conclusion : Non-significant

→ Days without social media does not predict exercise frequency.

RQ3: SLEEP QUALITY BY SOCIAL MEDIA PLATFORM AND AGE



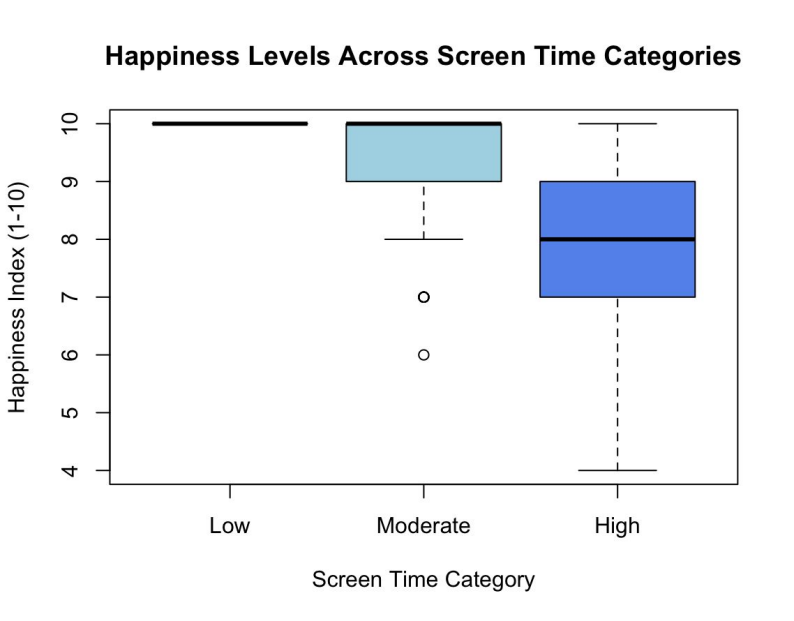


- **Independent variables:** Most used social media platform and age
- **Dependent variable:** Sleep quality (1-10)
- Used a Type II ANOVA test (no significant interaction)
- Results
  - Platform: f=1.9851, p=0.07939
  - Age: f=1.2037, p=0.27312

Conclusion: Since both p-values > 0.05, most used social media platform and age do not have a significant relationship with sleep quality scores.

RQ4: HAPPINESS LEVELS BY SCREEN TIME

- **Independent Variable**: Daily screen time in hours (Low: 0-2 h, Moderate: 2-5 h, High 5+h)
- **Dependent Variable**: Happiness levels (1-10)
- Grouped screen time into low, moderate, and high users then used an ANOVA test to see if happiness levels differs between the different groups. Using a tukey test, we can conclude which groups are different from each other.
- **ANOVA results**: F=112, p<0.001
- **Highly significant** → there is a difference in happiness levels across the 3 screen time levels
- **Tukey results**: High screen time users have significantly lower happiness than both moderate and low screen time users ($p < 0.001$), but there's no significant difference between low and moderate users.
- Boxplot shows happiness levels among the different screen time levels



OVERALL FINDINGS

- Most variables showed no significant relationships, suggesting social media platform choice, days offline, and age do not strongly predict mental health outcomes
- **One variable did show a strong effect:** daily screen time meaningfully impacts happiness
- **Stress Level vs. Social Media Platform**
  - Platform choice does not influence stress
- **Days Without Social Media vs. Exercise Frequency**
  - Taking breaks from social media does not predict how often people exercise
- **Platform + Age vs. Sleep Quality**
  - Neither age nor the platform someone uses most has a measurable effect on sleep quality
- **Screen Time Level vs. Happiness**
  - High screen time users (5+ hrs) have much lower happiness than low or moderate users
  - Low vs. moderate users do not differ
  - Excessive screen time is the only factor strongly tied to reduced happiness
- Screen time amount—not platform, age, or time offline—shows the strongest and only significant relationship with mental health

FUTURE EXPERIMENT

- Use a longitudinal design to track the same individuals over time
- Determine whether reducing screen time causes improvements in happiness or if unhappy people simply use screens more (reverse causation)
- Examine which **types** of screen time matter most
- Differentiate between social media use, work/study screen time, gaming, and passive content consumption
- Consider that not all forms of screen time may have the same impact on overall well-being

FUTURE IMPLICATIONS

- **Excessive screen time (5+ hours daily) correlates with sufficiently lower happiness levels** — a 1.7-2.3 point decrease that represents a medically meaningful difference in overall well-being
- **Public health recommendations may be needed** — organizations and educational institutions should consider establishing daily screen time limits for adults, similar to existing children's guidelines
- **Mental health screening should include screen time** — professionals treating depression or anxiety should start to assess patients' screen time habits, and digital detox interventions could become standard treatment components
- **Tech companies have a responsibility to discourage excessive use** — usage tracking and time limits should become industry standards rather than optional features that most users ignore
- **Moderate usage (2-5 hours) appears safe** — this range doesn't significantly differ from low usage, suggesting a potential "safe zone" where individuals can maintain healthier relationships with technology
- **Broader cultural shifts may be necessary** — society should reconsider norms around constant connectivity, "hustle culture" demanding 24/7 availability, and expectations of immediate digital responses, as our increasingly digital lifestyle appears to carry a psychological cost

SOURCES

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